



Comparative Effects of Waste-Derived Protein Ingredients on Growth, Feed Utilization and Health of *Anabas testudineus* During Winter

Md Aklakur¹, Md Imran Shah¹, Md Amjad¹, Udipta Roy¹, Potluri Sai Kishore², Sangeeta Mandal², Barkha Asthana¹

¹ Regional Research & Training Centre, ICAR-Central Institute of Fisheries Education, Motipur, Muzaffarpur, Bihar 843111, India

² ICAR-Central Institute of Fisheries Education, Versova, Andheri West, Mumbai 400061, India

ARTICLE INFO:

Research Article

Keywords: *Anabas testudineus*; Winter rearing; Animal waste; Growth performance; Haemato-biochemical; Enzymes

ABSTRACT

An experiment was conducted to evaluate the effects of various animal wastes as minor ingredients in growth, feed utilisation, and overall health status of *Anabas testudineus* in winter. In winter, fish productivity declines rapidly due to reduced metabolic activity, declined natural productivity, and stress. Utilisation of selected animal wastes may enhance aquaculture productivity by enhancing feed attractability and intake due to their free amino acids, soluble peptides, and flavour compounds that stimulate olfactory and gustatory response in fishes. The waste products: fish waste digest (FWD), biofloc meal (BFM), chicken waste hydrolysate (CWH), shrimp head meal (SHM), and beef liver meal (BLM) were supplemented at 5% inclusion rate in iso-nitrogenous and iso-lipid diet formulation and compared with control feeds. The fishes with initial weight of 1.5–1.6 g were stocked at 400/m³ density and cultured for 90 days. Throughout the experiment, the temperature ranged between 10.3 and 18.1 °C and all other water quality parameters were maintained at appropriate level. The growth parameters indicated significantly higher ($p < 0.05$) percentage weight gain ($361.45 \pm 3.50\%$), specific growth rate ($0.66 \pm 0.01\%$), average daily growth (1.24 ± 0.01 g), survivability ($73.33 \pm 6.01\%$), and improved feed conversion ratio (1.25 ± 0.01) in the BFM fed group. Similarly, the haemato-biochemical parameters indicated significantly higher haemoglobin (10.67 ± 0.26 g/dl), haematocrit ($4.80 \pm 0.15\%$), total erythrocyte count ($34.83 \pm 0.09 \times 10^6$ cells/mm³), total protein (3.63 ± 0.09 g/dl), albumin (0.94 ± 0.03 g/dl), globulin (2.69 ± 0.10 g/dl), and A/G ratio (0.35 ± 0.02) in BFM group. Inflammatory markers, digestive enzymes, and antioxidant enzyme levels demonstrated improved physiological responses in BFM group compared with control and other supplemented groups, supporting overall growth and haematological parameters of experimental animals. Whole body proximate indices suggest significantly higher lipid and ash content in control group (C) compared to other supplemented groups and however, crude protein content in BFM group compared with other supplemented groups. Therefore, supplementation of BFM at 5% inclusion rate enhances overall physiological welfare and survivability of *Anabas testudineus* cultured during winter.

1. Introduction

The rapid expansion of the aquaculture industry has highlighted significant environmental and economic challenges, particularly concerning the volume, sustainability, and supply potential of the traditional aquafeed production (Thazeem et al. 2022). Typically, commercial diets for carnivorous and omnivorous fish rely heavily on fishmeal and fish oil sourced from the wild-caught forage fish (Majluf et al. 2024). This reliance is inherently unsustainable, as it depends on the stability of global marine fisheries, which are subject to overfishing and decreasing catch rates (SAHIN 2025). The fluctuations in fishmeal supply and increasing costs have created a compelling need for identifying sustainable, viable, and economically feasible alternative protein sources and novel feed ingredients (Gasco et al. 2018; Nandi et al. 2023; Shimul et al. 2024; Roslan et al. 2025). These alternatives include plant-based proteins such as soybean, corn, and peas, which remain the most common substitutes (Kurek et al. 2022; Tay et al. 2023; Banach et al. 2023). Other promising options include insect meals, such as black soldier fly larvae, as well as microbial biomass derived from single-cell organisms, algae, and yeast, which are increasingly gaining attention (Hadi and Brightwell 2021; Pilmer 2025). Incorporating these ingredients in the aquaculture diets can help reduce feed costs and support a circular economy by valorising waste products (Colombo et al. 2023). However, widespread adoption of these alternative proteins faces notable challenges, primarily due to their potential negative impacts on feed palatability (Mitra 2021). Some of these ingredients may contain certain off-flavours that can reduce feed consumption, which in turn can compromise both water quality and fish growth performance. Consequently, the inclusion of dietary attractants can be critical in the modern aquafeed formulations (Conti et al. 2025).

Dietary attractants, also known as feeding stimulants, are substances that are added to aquafeeds to improve their flavour and stimulate feed intake, particularly in the diets where conventional marine-derived ingredients are reduced or substituted (Jannathulla et al. 2021; He et al. 2022). They facilitate appetite stimulation, promote gastric secretions, improve digestion, and enhance nutrient absorption, leading to better feed intake and nutrient assimilation in fish (Li et al. 2025). Several waste-derived animal proteins serve as both protein sources and feed attractants (Boboua et al. 2024). These ingredients are valued for their high protein content, rich amino acid profiles, and superior palatability compared to many plant-based alternatives (Boboua et al. 2024). Utilising waste-derived products as attractants can decrease dependence on expensive and unsustainable conventional feeds while also contributing to the valorisation of aquaculture waste streams (Afreem and Ucak 2020; Kari et al. 2023).

The *Anabas testudineus* (climbing perch) is a freshwater omnivore species that holds significant commercial and medicinal value across Southeast Asia (Morioka et al. 2009). This species is well-recognised for its relatively shorter life cycle, rapid growth, and resilience to challenging environmental conditions, making it a promising candidate for diversification within inland aquaculture systems (Debroy et al. 2022; Nandi et al. 2026). Despite its hardy nature due to its ability to tolerate extreme water quality and low oxygen levels, this species faces notable challenges in intensive aquaculture, particularly during the colder winter months (Sulekha and Mercy 2021). As a poikilothermic organism, its physiological and behavioural functions are significantly affected by fluctuations in water temperature, due to its inability to regulate its body temperature independently. Research indicates that during winter, especially in December and January, *A. testudineus* becomes more susceptible to stress and diseases, attributable to unfavourable water temperatures and critically low dissolved oxygen levels (Sarma et al. 2010). During this period, the species is at increased vulnerability for diseases like epizootic ulcerative syndrome (EUS) and protozoan infections (Ahmed et al. 2007). Additionally, at lower temperatures, the fish tend to reduce feed consumption due to decreased metabolic activity. This metabolic stress associated with fish's natural stress responses to cold conditions can lead to diminished feeding behaviour and stunted growth (Reid et al. 2022; Rojas et al. 2024). Farmers rearing *A. testudineus* often experience substantial financial losses due to reduced feed intake and heightened disease susceptibility during winter season (Ahmed et al. 2007). Therefore, the development of effective nutritional strategies is essential to support the increased energy requirements during cold stress and enhance disease resistance, ultimately lessening the negative effects associated with winter rearing. Although previous studies have investigated both conventional and alternative feed attractants in conventional culture systems, there remains a lack of comparative data on the efficacy of various waste-derived animal proteins, particularly for a high-value species like *Anabas testudineus* during its winter rearing. Although *Anabas testudineus* is recognised as a climate-resilient species, suitable for intensified culture, its performance under low temperature condition remains suboptimal due to poor feeding response. In this context, a cost-effective, attractant-enriched diet may represent a promising strategy to enhance feed actability thereby promoting overall growth of the fishes. Therefore, this study aims to evaluate the effects of dietary inclusion of fish waste digest, biofloc meal,

* Corresponding author: Udipta Roy, Senior Scientists and OIC, Motipur ICAR-CIFE, RRTC, Motipur, Muzaffarpur, Bihar 843111, India.

Email address: udiptaroy@cife.edu.i

DOI: <https://doi.org/10.1007/xxxxx.xxxxx>

Received 1 February 2026; Accepted 3 July 2026; Published online 22 July 2026.

This Aquaculture Scientific typeset version reproduces the supplied source article content in the journal's house format.

chicken waste hydrolysate, shrimp head meal, and beef liver meal on the growth performance and metabolic responses of juvenile *Anabas testudineus* during winter rearing conditions.

2. Materials and methods

2.1 Experimental location, animals, and diets

The 90-day feeding trial was conducted at the ICAR-Central Institute of Fisheries Education, RRTC Motipur, Bihar, during the winter season (November 2024 to January 2025) with an average temperature of 14.41 °C. The juveniles of *Anabas testudineus* were procured from the wild collection in local chaur (natural floodplains) and transported to the wet laboratory of ICAR-CIFE, RRTC Motipur, using oxygenated storage tanks (water storage tanks connected with aerators). They were acclimated in a 10,000 L FRP tank for 14 days and fed with a commercial feed (32% crude protein ‘cp’) at 6% of their body weight twice a day. Six iso-nitrogenous and iso-lipid diets containing 32% cp and 6% lipid were formulated according to the experimental design. The control diet (C) had no attractant, while the remaining diets contained 5% attractants, including 5% fish waste digest (FWD), 5% biofloc meal (BFM), 5% chicken waste hydrolysate (CWH), 5% shrimp head meal (SHM), and 5% beef liver meal (BLM). These diets were prepared through the following steps: the ingredients were initially ground and sieved. All ingredients were then weighed and thoroughly mixed as per to the formulation (Table 1). Then, the mixture was steamed for 20 min and allowed to cool at room temperature. After cooling, the remaining ingredients, including oil and vitamin-mineral mixtures, were added to the dough. The dough was pressed through a mechanical pelletiser fitted with a 1 mm diameter die to make the pellets that were air-dried, packed, and stored appropriately.

Table 1 Formulations sheet and feed proximate in the 90 days winter rearing of *Anabas testudineus*.

Ingredients (g 100 ⁻¹)	Control	FWD	BFM	CWH	SHM	BLM
Fish meal	5.00	5.00	5.00	5.00	5.00	5.00
Soybean meal	37.00	39.00	37.00	29.00	35.00	37.00
Ground nut oil cake	14.00	13.85	13.00	11.00	10.00	15.00
Mustard oil cake	11.00	10.42	11.00	13.00	12.00	9.00
Wheat flour	11.00	8.00	9.00	13.00	11.00	9.00
Rice bran	11.00	8.00	9.00	13.00	11.00	9.00
Corn flour	4.97	4.70	4.97	4.97	4.97	4.97
Sunflower oil	0.50	0.50	0.50	0.50	0.50	0.50
Groundnut oil	0.50	0.50	0.50	0.50	0.50	0.50
Fish oil	2.00	2.00	2.00	2.00	2.00	2.00
Vit-min mix	2.00	2.00	2.00	2.00	2.00	2.00
Carboxymethyl cellulose	1.00	1.00	1.00	1.00	1.00	1.00
Butylated hydroxytoluene	0.02	0.02	0.02	0.02	0.02	0.02
Vitamin C	0.01	0.01	0.01	0.01	0.01	0.01
FWD	0.00	5.00	0.00	0.00	0.00	0.00
BFM	0.00	0.00	5.00	0.00	0.00	0.00
CWH	0.00	0.00	0.00	5.00	0.00	0.00
SHM	0.00	0.00	0.00	0.00	5.00	0.00
BLM	0.00	0.00	0.00	0.00	0.00	5.00
Total	100.00	100.00	100.00	100.00	100.00	100.00
Proximate composition (g kg ⁻¹ , as fed basis)						
Moisture	9.77±0.04	9.84±0.02	9.76±0.03	9.71±0.06	9.89±0.04	9.67±0.04
Crude protein	32.14±0.02	32.41±0.03	32.40±0.03	32.44±0.02	32.40±0.03	32.42±0.07
Ether extract	6.19±0.05	6.27±0.04	6.12±0.01	6.36±0.01	6.18±0.05	6.26±0.03
Total ash	7.22±0.02	7.29±0.05	7.52±0.02	7.18±0.04	7.41±0.03	7.20±0.06
Crude fibre	5.47±0.04	5.36±0.01	5.83±0.01	5.73±0.02	5.89±0.02	5.40±0.03
Nitrogen free extract	48.98±0.12	48.67±0.12	48.13±0.08	48.29±0.09	48.12±0.12	48.72±0.17

Treatments—C, 0% additives, FWD 5% fish waste digest, BFM 5% floc meal, CWH 5% chicken waste hydrolysate, SHM 5% shrimp head meal, BLM 5% beef liver meal. Composition of the vitamin mineral mix (quantity/kg): vitamin A, 7,00,000 I.U.; vitamin D3, 70,000 I.U.; vitamin E, 250 mg; cobalt, 150 mg; copper, 1200 mg; iodine, 325 mg; iron, 1500 mg; magnesium, 6000 mg; potassium, 100 mg; sodium, 5.9 mg; manganese, 1500 mg; sulphur, 0.72%; zinc, 9600 mg; DL-methionine, 1000 mg; calcium, 25.5%; phosphorus, 12.75%. Data are expressed as mean ± S.E., n=3.

2.2 Preparation of minor ingredients or attractants

In this experiment, we have used 5 different meals as minor ingredient which is inoculated at 5% level, i.e. fish waste digest (FWD), biofloc meal (BFM), chicken waste hydrolysate (CWH), shrimp head meal (SHM), and beef liver meal (BLM). The fish waste digest and chicken waste hydrolysate were prepared using standard method as reported by Laishram et al. (2023). The biofloc meal was prepared using the protocol reported by Kheti et al. (2017). The commercial grade shrimp head meal was purchased from Amalgam Nutrients and Feed Limited (Cochin, India). The beef liver meal was prepared by grinding freshly procured beef liver, sterilising and cooking at 120 °C, drying at 60 °C then pulverising it into powder and storing at 4 °C (Boboua et al. 2024).

2.3 Experimental procedure

The experiment was conducted in the closed circular PVC tanks (100 L capacity) filled with 50 L freshwater. A completely randomised design (CRD) was followed consisting of six treatments with three replications, such as control (C) and the remaining treatments fed with the diet consisting of 5% attractants including FWD—fish waste digest, BFM—biofloc meal, CWH—chicken waste hydrolysate, SHM—shrimp head meal, and BLM—beef liver meal. After the initial acclimation, 360 healthy juveniles of *Anabas testudineus* (weight range of 1.5–1.6 g) were randomly distributed into 18 experimental tanks (20 juveniles per tank) at a stocking density of 400 juveniles per cubic metre. The formulated diet consisting of 32% cp and 6% lipid was fed to all the experimental fish twice daily (06.00 a.m., 07.00 p.m.) at 6% of their body weight during the entire experimental duration. During this 90-day experimental period, minimal water exchange was done by replenishing only 10% water volume per month from each tank to compensate for the evaporation loss. No aeration was provided throughout the experimental period.

2.4 Physico-chemical water quality parameters

The physico-chemical water parameters, including temperature (°C), pH, dissolved oxygen (mg/L), alkalinity (mg/L), total ammonia nitrogen (mg/L), nitrite (mg/L), and nitrate (mg/L), were measured weekly throughout the experimental period in accordance with APHA (1999) standard procedures (Table 2).

Table 2 Physico-chemical parameters optimum range of water in the 90 days winter rearing of *Anabas testudineus*.

Treatments	Control	FWD	BFM	CWH	SHM	BLM
Temperature (°C)	10.8–17.5	10.3–17.1	11–18.1	10.6–17.96	11.8–17.6	11.7–18.5
pH	7.4–8.4	7.8–8.2	8–8.4	7.4–8.2	7.4–8.2	8–8.4
TAN (mg/L)	0.25–1.50	0.25–1.65	0.30–1.25	0.15–1.55	0.25–1.35	0.30–1.25
Nitrite (mg/L)	0.01–0.35	0.01–0.85	0.01–0.55	0.01–0.48	0.01–0.62	0.01–0.50
Nitrate (mg/L)	0.01–1.86	0.01–2.33	0.01–2.06	0.01–2.45	0.01–2.68	0.01–1.93
DO (mg/L)	3.65–5.68	2.98–4.78	3.05–5.25	2.86–4.88	2.92–5.12	3.00–4.81
Alkalinity (mg/L)	128–214	110–221	102–204	114–198	120–214	125–202

Data expressed in range. Treatments—C, 0% additives, FWD 5% fish waste digest, BFM 5% biofloc meal, CWH 5% chicken waste hydrolysate, SHM 5% shrimp head meal, BLM 5% beef liver meal.

2.5 Sampling

The sampling was done at the end of the 90-day feeding trial and fish were enumerated to determine the survivability rate. The total weight of all fish in each treatment group was recorded to determine growth performance indices. Later, five fishes were randomly selected from each tank (15 per treatment) and anaesthetised using clove oil immersion at a concentration of 50 μ L/L. Blood samples were obtained by puncturing the caudal vein using 22 gauge needles and divided into two aliquots. One aliquot was immediately transferred into ethylenediaminetetraacetic acid (EDTA)-coated tubes and gently inverted to prevent coagulation for haematological analysis. The second aliquot was collected in tubes without anticoagulant, allowed to clot at room temperature, and subsequently centrifuged at RCF 1500 g at 4 °C for 10 min to separate the serum for biochemical analysis (Aysun and Karataş 2022). For tissue collection, the fish was anaesthetised using clove oil at a concentration of 50 μ L/L until the equilibrium and reflex responses is lost followed by euthanasia, in compliance with animal welfare guidelines. The liver, intestine, gill, and muscle tissue samples were collected from the fish immediately for enzymatic analysis. The five-fish sample (blood and tissue) from one tank was pooled to obtain one representative from each tank (n=3). These tissue samples were homogenised in chilled (0.25 M) sucrose solution using a mechanical tissue homogeniser and centrifuged at RCF 2500 g at 4 °C for 15 min (Xu et al. 2025). The supernatant obtained was stored at –20 °C for subsequent analysis. The total protein content of the separated supernatant was determined (Lowry et al. 1951).

2.6 Growth performance, feed utilisation parameters, and survivability

The growth performance of fish was monitored at regular sampling intervals and evaluated using the following growth parameters: final body weight (FBW), weight gain (WG), percentage weight gain (PWG), specific growth rate (SGR), and average daily growth (ADG). Additionally, feed utilisation was assessed using feed conversion ratio (FCR), feed efficiency ratio (FER), and protein efficiency ratio (PER) for all treatments. The survivability rate (SR) was also determined.

$$\begin{aligned}
 \text{WG} &= \text{Final body wt (g)} - \text{initial body wt (g)}. \\
 \text{WG(\%)} &= \frac{\text{Final body wt (g)} - \text{initial body weight (g)}}{\text{initial body weight (g)}}. \\
 \text{SGR (\%/day)} &= \frac{\ln(\text{final body wt in g}) - \ln(\text{initial body wt in g})}{\text{Number of days}} \times 100. \\
 \text{ADG (g/day)} &= \frac{\text{Final weight (g)} - \text{Initial weight (g)}}{\text{Total number of days of culture}}. \\
 \text{FCR} &= \frac{\text{Dry feed intake (g)}}{\text{body weight gain (g)}}. \\
 \text{FER} &= \frac{\text{Body weight gain (g)}}{\text{Feed intake (g)}}. \\
 \text{PER} &= \frac{\text{Body weight gain (g)}}{\text{intake of protein (dry matter in g)}}. \\
 \text{Survivability(\%)} &= \frac{\text{Number of fish harvested} - \text{Number of fish stocked}}{\text{Number of fish stocked}} \times 100.
 \end{aligned}$$

2.7 Proximate composition analysis of the experimental feeds

The proximate compositions of all the experimental feeds were determined according to the standard procedures (AOAC 1995) on the dry matter basis. The moisture content was analysed by drying the samples at 105 °C in a hot-air oven. The crude protein content was determined by the micro-Kjeldahl method (Tulin, India). The Soxhlet extraction method determined the total lipid content (Sox 6, Tulin, India). The crude fibre content was determined by acid–alkali digestion using automatic

crude fibre analysis system (FRB-6, Tulin, India), and the total ash content was determined by using a muffle furnace at 550 °C for 6 h. The following standard formula were used to calculate moisture, crude protein, total lipid, and total ash.

$$\begin{aligned}\text{Moisture (\%)} &= \frac{\text{sample weight before drying} - \text{sample weight after drying}}{\text{sample weight before drying}} \times 100. \\ \text{Crude protein (\%)} &= \text{Total nitrogen content (\%)} \times 6.25. \\ \text{Ether extract (\%)} &= \frac{\text{Wt. of dried receiver flask} + \text{extracted lipid} - \text{Wt. of dried receiver flask}}{\text{Weight of dried sample}} \times 100. \\ \text{Total Ash (\%)} &= \frac{\text{ash weight}}{\text{dry sample weight}} \times 100. \\ \text{Nitrogen free extract} &= 100 - (\text{crude protein} + \text{ether extract} + \text{total ash} + \text{crude fibre}).\end{aligned}$$

2.8 Haematological and serum biochemical parameters

Blood samples were collected for haematological parameters to assess haemoglobin content, haematocrit values, total erythrocyte count (RBC), and total leukocyte count (WBC). The haemoglobin content was estimated through the cyanmethemoglobin method using Drabkin's fluid. Twenty microlitres of blood was mixed with 5 ml of Drabkin's working solution and the absorbance was measured at 540 nm. RBC and WBC were estimated by counting the total number of cells in a haemocytometer. Serum biochemical parameters were also analysed using commercially available ERBA kits, including serum glucose, serum total protein, albumin, serum glutamic-oxaloacetic transaminase (SGOT), and serum glutamic-pyruvic transaminase (SGPT). Globulin was calculated by deducting the albumin values from the total serum protein. A/G ratio was calculated by in-between albumin values by globulin values.

2.9 Digestive and antioxidant enzyme analysis

Protease activity in the intestinal tissue homogenate was measured using the Bradford assay (Bradford 1976). Lipase activity was determined via the titrimetric method described by Cherry and Crandall (1932), and amylase activity was assessed using the DNS (3,5-dinitrosalicylic acid) assay as outlined by Rick and Stegbauer (1974). Additionally, the activities of antioxidative enzyme of both liver and gill samples were analysed: superoxide dismutase (SOD) activity was evaluated following the method by Misra and Fridovich (1972), while catalase activity was determined according to the procedure described by Takahara et al. (1960).

2.10 Statistical analysis

The data analysis was done in Statistical Package for the Social Sciences (SPSS), version 26.0 for Windows, by applying one-way analysis of variance (ANOVA) and expressed in mean±standard error. Duncan's multiple range test was used for the post hoc analysis at the 5% probability ($p < 0.05$). The normality of the dataset was tested using Kolmogorov–Smirnov and Shapiro–Wilk method.

3. Results

3.1 Physico-chemical water quality parameters

During the 90-day experimental trial, the water quality parameters were monitored to be within the ideal ranges for *Anabas testudineus* (Table 2). Across all treatment groups, the temperature and pH values ranged from 10.3 to 18.5 °C and 7.4 to 8.4, respectively. The values ranged from 0.15 to 1.65, 0.01 to 0.85, and 0.01 to 2.68 for total ammonia nitrogen (TAN), nitrite, and nitrate, respectively. Dissolved oxygen (DO) levels were maintained between 2.86 and 5.68 mg/L, while alkalinity varied from 102 to 221 mg/L.

3.2 Growth performance, feed utilisation parameters, and survivability

The 90-day feeding trial results showed a significant effect ($p < 0.05$) of dietary attractants on the growth performance and feed utilisation of juvenile *A. testudineus* (Table 3). All diets supplemented with attractants significantly improved ($p < 0.05$) growth and feed performance compared to the control group. The biofloc meal (BFM) group yielded the most improvement across all growth and feed utilisation parameters, followed by diets containing chicken waste hydrolysate (CWH), shrimp head meal (SHM), fish waste digest (FWD), beef liver meal (BLM), and the control diet. The BFM group recorded the highest final body weight (143.01 g), weight gain (112.01 g), and specific growth rate (0.664%), all of which were significantly greater ($p < 0.05$) than those of the other groups. The lowest FCR of 1.25 and the highest FER and PER of 0.803 and 2.23, respectively, were noticed in the BFM group (Table 4). The control group displayed the highest FCR of 2.60 and lowest FER and PER of 0.386 and 1.07, respectively. While the survivability rates did not differ significantly ($p > 0.05$) among the treatment groups, the BFM and FWD groups exhibited highest survivability.

Table 3 The effect of dietary supplementation with animal-based protein on growth performance of *Anabas testudineus* after 90 days of winter rearing.

Treatments	IW (g)	FW (g)	WG (g)	WG (%)	SGR (%)	ADG (g/day)
C	1.63±0.03	4.47±0.11 ^a	2.84±0.09 ^a	173.65±5.72 ^a	1.01±0.02 ^a	0.63±0.02 ^a
FWD	1.57±0.02	5.16±0.07 ^b	3.59±0.07 ^b	229.17±5.58 ^b	1.19±0.02 ^b	0.80±0.02 ^b
BFM	1.55±0.03	7.15±0.08 ^d	5.60±0.05 ^d	361.45±3.50 ^d	1.53±0.01 ^d	1.24±0.01 ^d
CWH	1.63±0.03	6.23±0.22 ^c	4.60±0.19 ^c	281.38±8.50 ^c	1.34±0.02 ^c	1.02±0.04 ^c
SHM	1.62±0.04	6.04±0.07 ^c	4.43±0.08 ^c	274.32±10.73 ^c	1.32±0.03 ^c	0.98±0.02 ^c
BLM	1.57±0.04	5.10±0.11 ^b	3.54±0.09 ^b	225.90±7.54 ^b	1.18±0.02 ^b	0.79±0.02 ^b
p value	0.379	0.001	0.001	0.001	0.001	0.001

Bold digits indicate highest value. Data expressed as mean±S.E., n=3, mean values in the same column with different superscripts differ significantly ($p < 0.05$). IW, initial weight; FW, final weight; WG, weight gain; WG%, weight gain percentage; SGR, specific growth rate (%/body); ADG, average daily growth. Normality of dataset was tested using Kolmogorov–Smirnov and Shapiro–Wilk method.

3.3 Haematological and serum biochemical parameters

All haematological and serum biochemical parameters exhibited significant differences across the treatment groups. The BFM group demonstrated the highest levels of haemoglobin, haematocrit, red blood cell counts, and the lowest white blood cell count compared to all other treatments (Table 5). Conversely, the control group showed an exactly opposite trend relative to the BFM group. Among the serum biochemical parameters, the BFM group showed the lowest glucose levels and the highest total protein, albumin, globulin, and A/G ratio values (Table 6). Additionally, levels of SGOT and SGPT were also lowest in the BFM group and highest in the control group.

Table 4 Feed performance and survivability of *Anabas testudineus* after 90 days of winter rearing.

Treatments	FCR	FER	PER	SR%
C	2.60±0.08 ^d	0.386±0.01 ^a	1.07±0.04 ^a	30.00±12.58
FWD	1.97±0.05 ^c	0.509±0.01 ^b	1.41±0.03 ^b	73.33±3.33
BFM	1.25±0.01 ^a	0.803±0.01 ^d	2.23±0.02 ^d	73.33±6.01
CWH	1.60±0.05 ^b	0.625±0.02 ^c	1.74±0.05 ^c	66.67±9.28
SHM	1.65±0.07 ^b	0.610±0.02 ^c	1.69±0.07 ^c	56.67±25.87
BLM	2.00±0.06 ^c	0.502±0.02 ^b	1.39±0.05 ^b	60.00±27.54
p value	0.001	0.001	0.001	0.503

Bold digits indicate significant value. Data are expressed as mean±S.E., n=5. FCR, feed conversion ratio; FER, feed efficiency ratio; PER, protein efficiency ratio; SR%, percentage survivability. Normality of the dataset was tested using Kolmogorov–Smirnov and Shapiro–Wilk method.

Table 5 Blood parameters in the 90 days winter rearing of *Anabas testudineus*.

Treat	Hb (g/dl)	Hct (%)	TEC (10 ⁶ cells/cumm)	TLC (10 ³ cells/cumm)
C	8.33±0.12 ^a	2.33±0.23 ^a	29.40±0.38 ^a	146.10±0.42 ^e
FWD	9.43±0.20 ^{b,c}	3.27±0.12 ^b	31.60±0.23 ^c	143.77±0.15 ^{c,d}
BFM	10.67±0.26 ^d	4.80±0.15 ^e	34.83±0.09 ^e	139.23±0.47 ^a
CWH	9.87±0.12 ^c	3.77±0.15 ^c	34.23±0.15 ^e	142.13±0.23 ^b
SHM	9.30±0.12 ^b	4.30±0.12 ^d	32.97±0.15 ^d	143.20±0.15 ^c
BLM	8.57±0.12 ^a	2.73±0.12 ^a	30.53±0.29 ^b	144.37±0.47 ^d
p value	0.001	0.001	0.001	0.001

Bold digits indicate significant values. Data are expressed as mean±S.E., n=5. Hb, haemoglobin; Hct, haematocrit; TEC, total erythrocyte count; TLC, total leukocyte count. Normality of the dataset was tested using Kolmogorov–Smirnov and Shapiro–Wilk method.

Table 6 Serum biochemical parameters in the 90 days winter rearing of *Anabas testudineus*.

Treat	Glucose (mg/dl)	Total protein (gm/dl)	Albumin (gm/dl)	Globulin (gm/dl)	A/G	SGOT (IU/dl)	SGPT (IU/dl)
C	87.73±0.27 ^e	2.20±0.06 ^a	0.53±0.01 ^a	1.67±0.05 ^a	0.32±0.01 ^{b,c}	73.00±0.12 ^d	31.23±0.15 ^c
FWD	85.10±0.12 ^c	2.90±0.06 ^c	0.58±0.01 ^a	2.32±0.05 ^c	0.25±0.01 ^a	72.10±0.15 ^c	30.47±0.57 ^c
BFM	82.83±0.20 ^a	3.63±0.09 ^e	0.94±0.03 ^d	2.69±0.10 ^d	0.35±0.02 ^c	69.40±0.35 ^a	27.77±0.15 ^a
CWH	84.03±0.15 ^b	3.23±0.09 ^d	0.78±0.01 ^c	2.46±0.08 ^c	0.32±0.01 ^{b,c}	71.13±0.24 ^b	28.60±0.15 ^b
SHM	85.07±0.26 ^c	3.03±0.09 ^{c,d}	0.67±0.02 ^b	2.36±0.08 ^c	0.28±0.01 ^{a,b}	71.70±0.17 ^{b,c}	29.33±0.12 ^b
BLM	86.90±0.12 ^d	2.50±0.06 ^b	0.55±0.01 ^a	1.95±0.07 ^b	0.29±0.02 ^{a,b}	72.77±0.15 ^d	30.67±0.09 ^c
p value	0.001	0.001	0.001	0.001	0.003	0.001	0.001

Bold digits indicate significant values. Data are expressed as mean±S.E., n=5. SGOT, serum glutamic-oxaloacetic transaminase; SGPT, serum glutamate pyruvate transaminase. Normality of dataset was tested using Kolmogorov–Smirnov and Shapiro–Wilk method.

3.4 Digestive and antioxidant enzyme activity

All digestive and antioxidant enzyme activities varied significantly ($p < 0.05$) across the treatment groups (Table 7). The BFM group consistently exhibited the highest activity for lipase, amylase, protease, SOD, and CAT enzymes compared to other groups. In contrast, the control showed the lowest activities of all the digestive and antioxidant enzymes comparatively.

Table 7 Digestive enzymes and antioxidant enzyme in the 90 days winter rearing of *Anabas testudineus*.

Treatments	Lipase	Amylase	Protease	SOD Liver	SOD Gill	Catalase Liver	Catalase Gill
C	0.35±0.02 ^a	1.21±0.02 ^a	1.28±0.01 ^a	3.27±0.15 ^a	3.23±0.09 ^a	4.93±0.15 ^a	1.15±0.03 ^a
FWD	0.44±0.02 ^b	1.38±0.02 ^c	1.60±0.02 ^b	4.03±0.09 ^{ab}	4.03±0.09 ^b	5.80±0.12 ^c	1.37±0.01 ^c
BFM	0.65±0.03 ^d	1.66±0.03 ^e	2.40±0.06 ^e	4.60±0.12 ^b	4.53±0.15 ^c	7.20±0.15 ^e	1.65±0.03 ^e
CWH	0.57±0.01 ^c	1.47±0.01 ^d	2.10±0.06 ^d	4.43±0.15 ^b	4.43±0.09 ^c	6.53±0.15 ^d	1.55±0.03 ^d
SHM	0.49±0.02 ^b	1.41±0.02 ^{cd}	1.92±0.04 ^c	5.87±0.71 ^c	4.00±0.12 ^b	5.87±0.23 ^c	1.45±0.03 ^c
BLM	0.44±0.01 ^b	1.30±0.03 ^b	1.38±0.02 ^a	3.97±0.09 ^{ab}	3.73±0.09 ^b	5.43±0.19 ^b	1.27±0.01 ^b
p value	0.001	0.001	0.001	0.001	0.001	0.002	0.001

Bold digits indicate significant values. Data are expressed as mean±S.E., n=3. SOD, superoxide dismutase. Normality of dataset was tested using Kolmogorov–Smirnov and Shapiro–Wilk method.

4. Discussion

4.1 Physico-chemical parameters of water

The key factor in winter rearing is water temperature, as it influences activity levels, feed intake, and growth in cultured fish (Richard et al. 2016). Typically, suboptimal thermal conditions can limit the fish production in aquaculture (O’Gorman et al. 2016), often resulting in decreased metabolic activity and feed intake, which leads to stunted growth and lower yields (Tejaswini et al. 2024). In this study, water temperatures ranged from 10.3 to 18.5 °C across all treatment groups. These levels fall substantially below the optimal temperature range of 20 to 30 °C for *Anabas testudineus* culture which can be established as the potential physiological stressor during this study. The pH remained stable between 7.4 and 8.4, while dissolved oxygen (DO) levels ranged from 2.86 to 5.68 mg/L. The optimum range of pH for *A. testudineus* culture ranged from 6 to 9 (Widiyati et al. 2019). The DO levels above 3 mg/L are generally considered acceptable (Widiyati et al. 2019). Alkalinity levels varied between 102 and 221 mg/L, remaining within the optimal range of 50–300 mg/L for *A. testudineus* culture (Debroy et al. 2022). The TAN and nitrite levels were also maintained within acceptable limits for *A. testudineus* culture, which ranged from 0.15 to 1.65 mg/L and 0.01 to 0.85 mg/L, respectively (Tay et al. 2006). Hence, throughout the 90-day winter rearing period, all the physico-chemical properties of the water were maintained consistently to ensure stable environmental conditions across all treatment groups. Consequently, any observed differences in fish growth and physiological status can be directly attributed to the specific dietary supplements, rather than variations in the rearing environment.

4.2 Growth performance and feed utilisation

The results from the 90-day winter rearing period demonstrate a significant effect ($p < 0.05$) of dietary supplementation with animal-based attractants on growth parameters and feed performance in *Anabas testudineus* (Table 3). In addition to achieving the lowest feed conversion ratio (FCR), the biofloc meal (BFM) treatment group also reached the highest levels ($p < 0.05$) in final body weight (FBW), weight gain (WG), weight gain percentage (WG%), specific growth rate (SGR), average daily gain (ADG), feed efficiency ratio (FER), and protein efficiency ratio (PER). The fundamental principles of biofloc technology (BFT) can be primarily attributed to the superior performance observed in the biofloc meal (BFM) diet. A complex admixture of organic nutrients and microorganisms, including bacteria, algae, nematodes, copepods, and protozoa, makes up biofloc (McCusker et al. 2023). This microbial biomass supplements the formulated diet by serving as a nutrient-rich food source, containing essential amino acids, high levels of crude protein, vitamins, and minerals. Beyond serving as a source of palatable protein, the biofloc composition includes highly bioavailable nutrients and bioactive compounds such as β -1,3-glucan, poly- β -hydroxybutyrate (PHB), and various vitamins and minerals (Panigrahi et al. 2018). These nutrients can be particularly crucial for *A. testudineus*, as they can help mitigate the physiological stress of cold temperatures and increase disease resistance (Ahmed et al. 2007). Studies have demonstrated that dietary inclusion of biofloc meal can enhance growth rates, improve immunological responses, and improve metabolic and digestive enzyme activities across various aquatic species (Mahanand et al. 2013; Das et al. 2025). The chicken waste hydrolysate (CWH) and shrimp head meal (SHM) treatments were the next most efficient feed for growth and feed performance with comparable efficiency to each other, followed by fish waste digest (FWD) and beef liver meal (BLM) groups, respectively. Several studies have documented that animal-derived protein hydrolysates can improve growth performance and nutrient utilisation in fish (Alves et al. 2020; Li et al. 2022; Cai et al. 2025). The control group showed highest FCR that indicates the lowest feed utilisation efficiency. Variations among the attractants might be due to differences in ingredient quality and processing methods. Both CWH and FWD diets consist of protein hydrolysates composed of highly digestible, low-molecular-weight peptides that can improve feed palatability and nutrient uptake. The superior growth and feed efficiency observed with CWH compared to FWD might be due to the qualitative difference in their composition or nutrient bioavailability. Shrimp head meal is well-known for its high chitin and protein content. The comparable performance of the CWH and SHM treatments suggests that these specific supplements offer a more effective nutritional and functional profile than FWD and BLM, highlighting that the efficacy of animal-based protein sources is heavily influenced by their specific origin and preparation methods (Afreen and Ucak 2020; Boboua et al. 2024). Usually, winter rearing of carps is associated with low survivability rates, often due to factors such as reduced metabolic activity, decreased feeding, and stunted fish growth (Kristan et al. 2018). The survivability rates across the treatments showed no significant differences ($p > 0.05$), indicating that the attractants had no adverse impacts on fish survivability during the winter rearing period. Similar studies were carried out by Mahanand et al. (2013) and Fang et al. (2022) who have also reported that dietary attractants generally have no major negative impact on the fish survivability. Compared to the attractant-fed groups, the control group exhibited significantly lower survivability rates, suggesting that these attractants helped sustain appetite and metabolic activity in the fish, thereby promoting feed intake during the winter season.

4.3 Haematological and serum biochemical indicators

The haematological profile is a standard indicator used to assess fish health and physiological responses to endogenous or exogenous stressors in fish (Ahmed et al. 2020). All haematological parameters demonstrated significant differences ($p < 0.05$) among all treatment groups. The highest levels of haemoglobin (Hb), haematocrit (Hct), and red blood cell (RBC) count were observed in the BFM group compared to the control group. Elevated Hb, Hct, and RBC counts are indicative of increased oxygen-carrying capacity of blood, which is a vital for physiological adaptation to meet metabolic demands and suggests relatively good health and resilience, particularly under cold stress conditions (Sharma and Shukla 2021). In contrast, the white blood cell (WBC) counts were lowest in the BFM treatment, while the control had the highest WBC. This reflects reduced systemic inflammation and a healthier, less stressed immune system, as a lower WBC count can indicate decreased immune activation in fish (Reid et al. 2022). Biofloc meal is rich in microbial proteins, polysaccharides, and other bioactive compounds, and it is recognised as a comprehensive biostimulant that can effectively support immune and metabolic functions in stressed fish (Ahmad et al. 2017; Tasleem et al. 2024). Dietary inclusion of 4 to 8% biofloc meal has shown to enhance immunity and disease resistance in rohu (*Labeo rohita*) fingerlings (Kheti et al. 2017).

The serum biochemical analysis also showed significant differences ($p < 0.05$) across all treatment groups, further supporting the physiological benefits associated with the supplemented attractants. The BFM group had the lowest serum glucose levels, indicating reduced stress, efficient energy utilisation, and well-regulated energy metabolism (Polakof et al. 2012; Wang et al. 2024). The highest levels of total protein and albumin were also observed in this group, accompanied by the lowest levels of liver enzymes SGOT and SGPT, indicating an improved nutritional status and active protein synthesis compared to other treatments. Elevated levels of SGOT and SGPT are typically associated with physiological stress, cellular damage, or liver malfunction (Singh et al. 2016; Satkar et al. 2024). These findings indicate that the BFM diet provided protection against cellular damage which promoted overall health, even under the challenging cold winter conditions as suggested by Yu et al. (2021).

4.4 Digestive and antioxidant enzyme activities

During the winter season, carp culture is greatly affected by the low temperatures, which lead to a marked decrease in feed intake and digestive enzyme activities (Afzal Khan et al. 2004; Tejaswini et al. 2024). In this study, significant differences ($p < 0.05$) were observed in the activities of all three digestive enzymes, lipase, amylase, and protease, across the treatment groups. The BFM group exhibited the highest activity for all three enzymes, followed by the CWH and SHM groups. The control and BLM groups demonstrated the lowest enzyme activities. These findings suggest that biofloc meal acted as superior attractant to help counteract reduced feed intake during cold conditions by stimulating appetite and enhancing digestive processes. Usually, higher feed intake and substrate availability leads to the increment in enzyme activity (Tejaswini et al. 2024). Microbial bioflocs are well-known to contain several exogenous and endogenous enzymes that greatly improves the nutrient digestibility in fish (Bakhshi et al. 2016). Additionally, several bioactive components of the biofloc including polyunsaturated fatty acids, cholesterol, and phospholipids may enhance the thermal tolerance under low temperature conditions (Hoffling et al. 2024). Similarly, the dietary inclusion of 0.05% compound feed attractants has enhanced the digestive enzyme activities in *P. monodon* under low-temperature stress (Liao et al. 2025). The consistent enhanced activity of digestive enzymes in all the dietary attractants indicates improved nutrient breakdown and absorption, which can be correlated to the superior growth and feed efficiency. Recent studies also reported a similar improvement in the digestive enzyme activities in fish following the addition of various dietary attractants (Li et al. 2023; Fang et al. 2022; Huang et al. 2025).

During winter rearing, the suboptimal water temperatures induce a state of metabolic stress and increased oxygen demand that leads to oxidative stress in fish (Deo et al. 2025; Satyakumar et al. 2025). Usually, the superoxide dismutase (SOD) and catalase (CAT) are considered the first line of defence in fish against oxidative stress, which works to neutralise harmful reactive oxygen species (Kishore et al. 2025). In comparison to the control and BLM groups, the BFM, CWH, and SHM groups exhibited significantly higher levels of SOD and CAT in both liver and gill tissues. These results suggest that diets incorporating these attractants fortified the endogenous antioxidant systems in fish, which mitigated cold-induced cellular damage and supported overall health. Various studies have suggested that apart from simple nutritional supplementation, the biofloc systems usually consists of functional and health-promoting properties that enhance antioxidant mechanism in carps (Adineh et al. 2019; Yu et al. 2020; Eljasik et al. 2022).

5. Conclusion

The findings demonstrate that various animal-based proteins, including fish waste digest, biofloc meal, chicken waste hydrolysate, shrimp head meal, and beef liver meal, can serve as effective feed attractants for *Anabas testudineus* during winter rearing. Among these, biofloc meal (BFM) was identified as the most effective dietary attractant compared to all other animal-based attractants used in this study. Even under challenging environmental conditions, the BFM diet showed significant improvements in growth performance, feed efficiency, and overall physiology or health of *A. testudineus*. Beyond growth, the BFM diet enhanced the digestive and antioxidant enzyme activities and contributed to improved haematological and serum profiles. This research suggests that incorporating such functional ingredients can be a viable strategy for maintaining a healthy and productive *A. testudineus* stock during the stressful periods of winter season.

Supplementary information

The online version contains supplementary material available at <https://doi.org/10.1007/s10499-026-02613-z>.

Acknowledgements

The authors are thankful to the Director and Vice-Chancellor of ICAR-CIFE, Mumbai, who have consistently provided facilitation and support at RRTC Motipur, ICAR-CIFE, which has been instrumental in the timely completion of this work.

Author contribution

Md Aklakur: Supervision, & conceptualization, execution of trial, methodology. Md Imran Shah: Draft, Formal analysis, Methodology, software analysis. Md Amjad: Investigation. Udipta Roy: Draft, Methodology, Investigation, Visualisation & supervision. Potluri Sai Kishore: Draft, reviewing & editing. Sangeeta Mandal: Reviewing & editing. Barkha Asthana: Reviewing & editing.

Funding

Not applicable.

Data availability

The datasets generated and/or analyzed during the current study are not publicly available as the research is conducted under ICAR-CIFE, RRTC, Motipur through a commercial biofloc fish farm, a government body which do not share the raw data to public forums, but it could be accessed from the corresponding author on reasonable and valid request.

Declarations

Ethics approval and consent to participate. All procedures were conducted in accordance with the Prevention of Cruelty to Animals Act, 1960, approved by CPCSEA (Committee for the Purpose of Control and Supervision of Experiments on Animals) and conducted under the registration number approved to the institute (ICAR-CIFE) with respect to registration number 524/GO/Re/S/02/CPCSEA. The research was conducted following all rules and regulations recommended as per the certificate. The ARRIVE guidelines have been followed in reporting the study.

Conflict of interest. The authors declare no competing interests.

Clinical trial number. Not applicable.

Artificial intelligence. Author(s) hereby declares that no generative AI technologies such as large language models (ChatGPT, etc.) and text generators have been used during writing or editing of this manuscript.

References

- Adineh H, Naderi M, Hamidi MK, Harsij M (2019) Biofloc technology improves growth, innate immune responses, oxidative status, and resistance to acute stress in common carp (*Cyprinus carpio*) under high stocking density. *Fish Shellfish Immunol* 95:440–448. <https://doi.org/10.1016/j.fsi.2019.10.057>
- Afreen M, Ucak I (2020) Fish processing wastes used as feed ingredient for animal feed and aquaculture feed. *J Surv Fish Sci* 6(2):55–64. <https://doi.org/10.18331/SFS2020.6.2.7>
- Afzal Khan M, Jafri AK, Chadha NK (2004) Growth and body composition of rohu, *Labeo rohita* (Hamilton), fed compound diet: winter feeding and rearing to marketable size. *J Appl Ichthyol* 20(4):265–270. <https://doi.org/10.1111/j.1439-0426.2004.00550.x>
- Ahmad I, Babitha Rani AM, Verma AK, Maqsood M (2017) Biofloc technology: an emerging avenue in aquatic animal healthcare and nutrition. *Aquac Int* 25(3):1215–1226. <https://doi.org/10.1007/s10499-016-0108-8>

- Ahmed I, Reshi QM, Fazio F (2020) The influence of the endogenous and exogenous factors on hematological parameters in different fish species: a review. *Aquac Int* 28(3):869–899. <https://doi.org/10.1007/s10499-019-00501-3>
- Ahmed GU, Dhar M, Khan MNA, Choi JS (2007) Investigation of diseases of Thai koi, *Anabas testudineus* (Bloch) from farming conditions in winter. *J Life Sci* 17(10):1309–1314
- Alves DRS, de Oliveira SR, Luczinski TG, Boscolo WR, Bittencourt F, Signor A, Detsch DT (2020) Attractability and palatability of liquid protein hydrolysates for Nile tilapia juveniles. *Aquac Res* 51(4):1681–1688. <https://doi.org/10.1111/are.14514>
- AOAC (1995) Official methods of analysis, 16th edn. Association of Official Analytical Chemists, Virginia, USA
- APHA (1999) Standard methods for the examination of water and wastewater, 20th edn. American Public Health Association, Washington DC, p 1268
- Aysun KOP, Karataş B (2022) Advances in aquaculture and fisheries research, 1st edn. IKSAD International Publishing House, Ankara, Türkiye. https://www.researchgate.net/profile/Boran-Karatas/publication/366064975_ADVANCES-IN-AQUACULTURE-AND-FISHERIES-RESEARCH/links/6390382ee42faa7e75a0f8bf/ADVANCES-IN-AQUACULTURE-AND-FISHERIES-RESEARCH.pdf. Accessed 2026
- Bakhshi F, Najde Gerami EH, Imani A, Sarvi Moghanlou K (2016) Effect of biofloc technology on growth performances, body composition and reduction of economic costs in intensive culture of common carp (*Cyprinus carpio*) juveniles. *J Vet Res* 71(2):163–9. In Persian with English Summary
- Banach JL, Van Der Berg JP, Kleter G, Van Bokhorst-van De Veen H, Bastiaan-Net S, Pouvreau L, Van Asselt ED (2023) Alternative proteins for meat and dairy replacers: food safety and future trends. *Crit Rev Food Sci Nutr* 63(32):11063–11080. <https://doi.org/10.1080/10408398.2022.2089625>
- Boboua SYB, We Q, Zhang L, Chen Y, Yu J, Chen P, Sun Y, Zheng T (2024) Valorization of animal waste proteins for agricultural, food production, and medicinal applications. *Front Sustain Food Syst* 8:1366333. <https://doi.org/10.3389/fsufs.2024.1366333>
- Bradford M (1976) Differential color change of a dye in response to various concentrations of protein. *Anal Biochem* 72:248–253
- Cai G, Li YuM, Huang M, Liu P, Tang X, Huang Q, Guo Z, Sun Y (2025) Dietary supplementation of an organic acid-based feed attractant in juvenile largemouth bass (*Micropterus salmoides*): effects on growth, morphohistology, and oxidative stress. *Fishes* 10(5):195. <https://doi.org/10.3390/fishes10050195>
- Cherry IS, Crandall LA Jr (1932) The specificity of pancreatic lipase: its appearance in the blood after pancreatic injury. *Am J Physiol Legacy Content* 100(2):266–273. <https://doi.org/10.1152/ajplegacy.1932.100.2.266>
- Colombo SM, Roy K, Mraz J, Wan AH, Davies SJ, Tibbetts SM, Øverland M, Francis DS, Rocker MM, Gasco L, Spencer E (2023) Towards achieving circularity and sustainability in feeds for farmed blue foods. *Rev Aquacult* 15(3):1115–1141. <https://doi.org/10.1111/raq.12766>
- Conti F, Zarantonello M, Cattaneo N, Antonucci M, Belfiore EA, Olivotto I (2025) Synthetic feed attractants in European seabass (*Dicentrarchus labrax*) culture: effects on growth, health, and appetite stimulation. *Animals (Basel)* 15(14):2060. <https://doi.org/10.3390/ani15142060>
- Das PC, Choudhary P, Ferosekhan S, Kumari R, Swain HS (2025) Influence of different stocking densities on growth performance, water quality, immunological responses, and physiological status of *Anabas testudineus* fingerlings reared in a biofloc-based production system. *Aquacult Int* 33(6):568. <https://doi.org/10.1007/s10499-025-02254-8>
- Debroy S, Chadha NK, Prakash S, Sawant PB, Hari Krishna V, Pathan MA, Haque R, Jana P, Roy U (2022) Effect of salinity on growth, survivability, haemato-biochemical and antioxidative status of *Anabas testudineus* (Bloch, 1792) juveniles reared in inland saline water. *Aquac Res* 53(18):6832–6845. <https://doi.org/10.1111/are.16149>
- Deo AD, Jayant M, Sahu NP, Shamna N, Rasal KD, Dhanalakshmi M, Kayalvizhi AA, Prashanth BR (2025) Dietary chromium enhances carbohydrate utilization and metabolic gene expression while alleviating cold stress in *Labeo rohita* fingerlings. *Animal Feed Science and Technology* 116419:116419. <https://doi.org/10.1016/j.anifeedsci.2025.116419>
- Eljasik P, Panicz R, Sobczak M, Sadowski J (2022) Key performance indicators of common carp (*Cyprinus carpio* L.) wintering in a pond and RAS under different feeding schemes. *Sustainability* 14(7):3724. <https://doi.org/10.3390/su14073724>
- Fang T, Li X, Wang J, Guan D, Sun H, Yun X (2022) Zhou J (2022) Effects of three compound attractants in plant protein diets on growth, immunity, and intestinal morphology of yellow river carp *Cyprinus carpio* var. *Aquac Nutr* 1:9510968. <https://doi.org/10.1155/2022/9510968>
- Gasco L, Gai F, Maricchiolo G, Genovese L, Ragonese S, Bottari T, Caruso G (2018) Fishmeal alternative protein sources for aquaculture feeds. *Feeds for the aquaculture sector*. Springer, Cham, pp 1–28. https://doi.org/10.1007/978-3-319-77941-6_1
- Hadi J, Brightwell G (2021) Safety of alternative proteins: technological, environmental and regulatory aspects of cultured meat, plant-based meat, insect protein and single-cell protein. *Foods* 10(6):1226. <https://doi.org/10.3390/foods10061226>
- He G, Chen X, Zeng Q, Zhu W, Chen Z, Tan B, Xie S (2022) Effects of compound feed attractants on growth performance, feed utilization, intestinal histology, protein synthesis, and immune response of white shrimp (*Litopenaeus vannamei*). *Animals* 12(19):2550. <https://doi.org/10.3390/ani12192550>
- Hoffling FB, Marquezi AS, Pinheiro I, Simon C, Rombenso AN, Seiffert WQ, Vieira FDN, Schleder DD (2024) *Aurantiochytrium* sp. meal as feed additive for Pacific white shrimp reared under low temperature and challenged by WSSV in association with thermal stress. *Fishes* 9(3):108. <https://doi.org/10.3390/fishes9030108>
- Huang W, Xu W, Yao C, Liu Y, Xu N, Yin Z, Miao Y, Mai K, Ai Q (2025) Compound nucleotides as feed attractants improve the survivability, growth, digestion, antioxidant capacity, and immunity of large yellow croaker (*Larimichthys crocea*) larvae. *Fish and Shellfish Immunology* 110435:110435. <https://doi.org/10.1016/j.fsi.2025.110435>
- Jannathulla R, Sravanthi O, Khan HI, Moomeen HS, Gomathi A, Dayal JS (2021) Chemoattractants: their essentiality and efficacy in shrimp aquaculture. *Indian J Fish* 68(1):151–159. <https://doi.org/10.21077/ijf.2021.68.1.95994-20>
- Kari ZA, Sukri SAM, Rusli ND, Mat K, Mahmud MB, Zakaria NNA, Wee W, Hamid NKA, Kabir MA, Ariff NSNA, Abidin SZ (2023) Recent advances, challenges, opportunities, product development and sustainability of main agricultural wastes for the aquaculture feed industry—a review. *Ann Anim Sci* 23(1):25–38. <https://doi.org/10.2478/aoas-2022-0082>
- Kheti B, Kamilya D, Choudhury J, Parhi J, Debbarma M, Singh ST (2017) Dietary microbial floc potentiates immune response, immune relevant gene expression and disease resistance in rohu, *Labeo rohita* (Hamilton, 1822) fingerlings. *Aquaculture* 468:501–507. <https://doi.org/10.1016/j.aquaculture.2016.11.018>
- Kishore S, Kumar P, Shamna N, Sardar P, Sahu NP, Patra O, Solomon J, Bhusare S (2025) Effect of dietary-supplemented *Centella asiatica* leaf extract on growth, nutrient utilization and physio-metabolic responses in gift fingerlings of different stocking densities reared in 10 ppt inland saline water. *Anim Feed Sci Technol* 327:116395. <https://doi.org/10.1016/j.anifeedsci.2025.116395>
- Kristan J, Blecha M, Polcar T (2018) Survivability and growth rates of juvenile grass carp *Ctenopharyngodon idella* overwintered in ponds and recirculating aquaculture systems including a comparison of production economics. *Turk J Fish Aquat Sci* 19(3):261–266. https://doi.org/10.4194/1303-2712-v19_3_09
- Kurek MA, Onopiuk A, Pogorzelska-Nowicka E, Szpicer A, Zalewska M, Półtorak A (2022) Novel protein sources for applications in meat-alternative products—insight and challenges. *Foods* 11(7):957. <https://doi.org/10.3390/foods11070957>
- Laishram M, Desai AS, Pathan DI, Pawase AS, Wasave SM (2023) Functional properties and proximate analysis of fish waste protein hydrolysate processed using enzymes. *Curr Res Nutr Food Sci J* 11(2):641–654. <https://doi.org/10.12944/CRNFSJ.11.2.16>
- Li W, Li E, Wang S, Liu J, Wang M, Wang X, Qin C, Qin J, Chen L (2025) Comparative effects of four feed attractants on growth, appetite, digestion and absorption in juvenile Chinese mitten crab (*Eriocheir sinensis*). *Aquaculture* 594:741441. <https://doi.org/10.1016/j.aquaculture.2024.741441>
- Li X, Fang T, Wang J, Wang Z, Guan D, Sun H, Yun X, Zhou J (2022) The efficiency of adding amino acid mixtures to a diet free of fishmeal and soybean meal as an attractant in yellow river carp (*Cyprinus carpio* var.). *Aquac Rep* 24:101189. <https://doi.org/10.1016/j.aqrep.2022.101189>
- Li Y, Wang YL, Zhang WJ, Han HJ, Wang Z, Wang JT, Sheng YC (2023) Effects of different kinds of compound attractant in plant protein diet on growth, immune and intestinal digestive enzyme activities of genetically improved farmed tilapia (*Oreochromis niloticus*). <https://doi.org/10.12418/CJAN2023.490>
- Liao Z, Chen M, Gu X, Li X, Zhang W, Ye T, Zhao W, Tan B, Niu J (2025) Enhancing shrimp survivability and stress response: the role of compound feed attractants in low-temperature conditions. *Aquaculture* 742957. <https://doi.org/10.1016/j.aquaculture.2025.742957>
- Lowry OH, Rosebrough NJ, Farr AL, Randall RJ (1951) Protein measurement with the Folin phenol reagent. *J Biol Chem* 193(1):265–275
- Mahanand SS, Moulick S, Srinivasa Rao P (2013) Optimum formulation of feed for rohu, *Labeo rohita* (Hamilton), with biofloc as a component. *Aquac Int* 21(2):347–360. <https://doi.org/10.1007/s10499-012-9557-x>
- Majluf P, Matthews K, Pauly D, Skerrett DJ, Palomares MLD (2024) A review of the global use of fishmeal and fish oil and the fish in:fish out metric. *Sci Adv* 10(42):eadn

5650. <https://doi.org/10.1126/sciadv.adn5650>

- McCusker S, Warberg MB, Davies SJ, Valente CDS, Johnson MP, Cooney R, Wan AH (2023) Biofloc technology as part of a sustainable aquaculture system: a review on the status and innovations for its expansion. *Aquac Fish* 3(4):331–352. <https://doi.org/10.1002/aff2.108>
- Misra HP, Fridovich I (1972) The role of superoxide anion in the autoxidation of epinephrine and a simple assay for superoxide dismutase. *J Biol Chem* 247(10):3170–3175
- Mitra A (2021) Thought of alternate aquafeed: conundrum in aquaculture sustainability? *Proc Zool Soc* 74:1–18. <https://doi.org/10.1007/s12595-020-00352-4>
- Morioka S, Ito S, Kitamura S, Vongvichith B (2009) Growth and morphological development of laboratory-reared larval and juvenile climbing perch *Anabas testudineus*. *Ichthyol Res* 56(2):162–171. <https://doi.org/10.1007/s10228-008-0081-y>
- Nandi SK, Hridoy MAAM, Sarkar S, Kari ZA, Siddik MA, Hasan AMN, Eissa ESH, Van Doan H, Reduan MFH, Kabir MA (2026) Zootechnical performance, disease resistance and economic efficiency of climbing perch, *Anabas testudineus* fed *Terminalia arjuna* bark-supplemented diets. *Aquaculture Reports* 46:103328. <https://doi.org/10.1016/j.aqrep.2025.103328>
- Nandi SK, Suma AY, Rashid A, Kabir MA, Goh KW, Abdul Kari Z, Van Doan H, Zakaria NNA, Khoo MI, Seong Wei L (2023) The potential of fermented water spinach meal as a fish meal replacement and the impacts on growth performance, reproduction, blood biochemistry and gut morphology of female stinging catfish (*Heteropneustes fossilis*). *Life* 13(1):176. <https://doi.org/10.3390/life13010176>
- O’Gorman EJ, Ólafsson ÓP, Demars BO, Friberg N, Guðbergsson G, Hannesdóttir ER, Jackson MC, Johansson LS, McLaughlin ÓB, Ólafsson JS, Woodward G (2016) Temperature effects on fish production across a natural thermal gradient. *Glob Chang Biol* 22(9):3206–3220. <https://doi.org/10.1111/gcb.13233>
- Panigrahi A, Saranya C, Sundaram M, Kannan SV, Das RR, Kumar RS, Rajesh P, Otta SK (2018) Carbon:nitrogen (C:N) ratio level variation influences microbial community of the system and growth as well as immunity of shrimp (*Litopenaeus vannamei*) in biofloc based culture system. *Fish Shellfish Immunol* 81:329–337. <https://doi.org/10.1016/j.fsi.2018.07.035>
- Pilmer LW (2025) Sustainable alternatives to fishmeal in yellowtail kingfish (*Seriola lalandi*) diets: evaluating single-cell proteins and black soldier fly meal for growth performance and health. Doctoral dissertation, Murdoch University. <https://s1lnk.com/CVbGE>. Accessed May 2026
- Polakof S, Panserat S, Soengas JL, Moon TW (2012) Glucose metabolism in fish: a review. *J Comp Physiol B* 182(8):1015–1045. <https://doi.org/10.1007/s00360-012-0658-7>
- Reid CH, Patrick PH, Rytwinski T, Taylor JJ, Willmore WG, Reesor B, Cooke SJ (2022) An updated review of cold shock and cold stress in fish. *J Fish Biol* 100(5):1102–1137. <https://doi.org/10.1111/jfb.15037>
- Richard N, Silva TS, Wulff T, Schrama D, Dias JP, Rodrigues PM, Conceição LE (2016) Nutritional mitigation of winter thermal stress in gilthead seabream: associated metabolic pathways and potential indicators of nutritional state. *J Proteomics* 142:1–14. <https://doi.org/10.1016/j.jprot.2016.04.037>
- Rick W, Stegbauer HP (1974) α -Amylase measurement of reducing groups. *Methods of enzymatic analysis*. Academic Press, pp 885–890. <https://doi.org/10.1016/B978-0-12-091302-2.50074-8>
- Rojas I, Caballero-Solares A, Vadboncoeur É, Sandrelli RM, Hall JR, Clow KA, Parrish CC, Rise ML, Swanson AK, Gamperl AK (2024) Prolonged cold exposure negatively impacts Atlantic salmon (*Salmo salar*) liver metabolism and function. *Biology (Basel)* 13(7):494. <https://doi.org/10.3390/biology13070494>
- Roslan NA, Kari ZA, Sukri SAM, Khoo MI, Sani NIM, Rashid R, Kabir MA, Nandi SK, Zakaria NNA, Ghosh K, Harun HC (2025) Fermented spent coffee ground in African catfish (*Clarias gariepinus*) diets: effects on growth performance, digestive enzyme, protein digestibility, amino acid profile, and immune-related gene. *Aquaculture* 603:742383. <https://doi.org/10.1016/j.aquaculture.2025.742383>
- Şahin T (2025) Ecological and geopolitical challenges in sustaining global fishmeal supply for aquaculture: challenges in sustaining global fishmeal supply for aquaculture. *Marine Reports (MAREP)* 4(1):55–67. <https://doi.org/10.5281/zenodo.15764050>
- Sarma K, Pal AK, Ayyappan S, Das T, Manush SM, Debnath D, Baruah K (2010) Acclimation of *Anabas testudineus* (Bloch) to three test temperatures influences thermal tolerance and oxygen consumption. *Fish Physiol Biochem* 36(1):85–90. <https://doi.org/10.1007/s10695-008-9293-3>
- Satkar SG, Bhusare S, Sahu A, Aitwar VS, Kunjir SN, Girkar MM (2024) Stress-induced hematological and serum alterations as biomarkers in fish health assessment. *Int J Adv Biochem Res* 8(2):416–420. <https://doi.org/10.33545/26174693.2>
- Satyakumar A, Varghese T, Nathaniel TP, Bhatt S, Vasanthakumaran K, Gupta S, Deo AD (2025) Elevated rearing temperatures increase dietary lipid requirements in GIFT tilapia, *Oreochromis niloticus* (Linnaeus, 1758). *J Environ Biol* 46(4):535–545. <https://doi.org/10.22438/jeb/46/4/MRN-5503>
- Sharma M, Shukla P (2021) Impact of temperature variation on haematological parameters in fish *Cyprinus carpio*. *J Entomol Zool Stud* 9(2):134–136. <https://11nq.com/eyBDL>. Accessed Apr 2026
- Shimul TJT, Kallem P, Nandi SK, Afrin S, Suma AY, Kari ZA, Sukri SAM, Khoo MI, Roslan NA, Guru A, Velayudhannair K (2024) Fermented aquatic weed meal (FAWM) as a protein source in Asian catfish *Clarias batrachus* diets: impacts on growth, blood chemistry profile, liver and gut morphology and economic efficiency. *Aquac Rep* 38:102305. <https://doi.org/10.1016/j.aqrep.2024.102305>
- Singh S, Rawat RS, Singh S, Sharma HN (2016) Studies on liver marker enzymes (SGOT and SGPT) of fish *Heteropneustes fossilis* (Bloch.) after Famfos intoxication. *J Adv Lab Res Biol* 7(4):99–102. <https://s1lnk.com/40GqJ>. Accessed May 2026
- Sulekha BT, Mercy TA (2021) Histopathological changes in the liver and kidney of *Anabas testudineus* exposed to sub-lethal concentrations of phosphamidon. *Uttar Pradesh J Zool* 42(24):814–820. <https://s1lnk.com/xVORi>
- Takahara S, Hamilton HB, Neel JV, Kobara TY, Ogura Y, Nishimura ET (1960) Hypocatalasemia: a new genetic carrier state. *J Clin Invest* 39(4):610–619. <https://doi.org/10.1172/JCI104075>
- Tasleem S, Alotaibi BS, Masud S, Habib SS, Acar Ü, Gualandi SC, Ullah M, Khan K, Fazio F (2024) Khayyam K (2024) Biofloc system with different carbon sources improved growth, haematology, nonspecific immunity, and resistivity against the *Aeromonas hydrophila* in common carp. *Cyprinus Carpio Aquaculture Research* 1:7652354. <https://doi.org/10.1155/2024/7652354>
- Tay W, Quek R, Lim J, Kaur B, Ponnalagu S, Henry CJ (2023) Plant-based alternative proteins—are they nutritionally more advantageous? *Eur J Clin Nutr* 77(11):1051–1060. <https://doi.org/10.1038/s41430-023-01328-1>
- Tay YL, Loong AM, Hiong KC, Lee SJ, Tng YY, Wee NL, Lee SM, Wong WP, Chew SF, Wilson JMI (2006) Active ammonia transport and excretory nitrogen metabolism in the climbing perch, *Anabas testudineus*, during 4 days of emersion or 10 minutes of forced exercise on land. *J Exp Biol* 209(22):4475–4489. <https://doi.org/10.1242/jeb.02557>
- Tejaswini K, Deo AD, Shamna N, Jayant M, Aklakur M, Annadurai R (2024) Effect of flavanone rich lemon peel extract on feed intake and growth of *Labeo rohita* (Hamilton, 1822) fingerlings reared at low temperature recirculatory aquaculture system. *Aquaculture* 584:740450. <https://doi.org/10.1016/j.aquaculture.2023.740450>
- Thazeem B, Umesh M, Sarojini S, Allwyn Vyas G, Adhithya Sankar S, Saphthami K, Suresh S, Stanly LM (2022) Developments in feeds in aquaculture sector: contemporary aspects. In: Balasubramanian B, Liu WC, Sattanathan G (eds) *Aquaculture science and engineering*. Springer, Singapore, pp 35–78. https://doi.org/10.1007/978-981-19-0817-0_2
- Wang A, Zhang R, Zhang X, Chen C, Gong Q, Wang L, Wang Y (2024) Effects of cold acclimation on serum biochemical parameters and metabolite profiles in *Schizothorax prenanti*. *BMC Genomics* 25(1):547. <https://doi.org/10.1186/s12864-024-10483-z>
- Widiyati A, Saputra A, Setiadi E (2019) Production performance and blood profile of climbing perch *Anabas testudineus* Bloch cultured in peat pond with different stocking densities. *Indones Aquacult J* 14(2):83–89
- Xu Q, Tang Q, Yi L, Gong Y, Shen Z, Yuan Y (2025) Selenium alleviates the effects of cadmium on growth performance, oxidative stress, cholesterol synthesis and ferroptosis of hybrid yellow catfish, *Pelteobagrus fulvidraco* female × *Pelteobagrus vachellimale*. *Comp Biochem Physiol C Toxicol Pharmacol*. <https://doi.org/10.1016/j.cbpc.2025.110290>
- Yu Z, Dai ZY, Li L, Qin GX, Wu LF (2021) Dietary supplementation with biofloc promotes growth, improves immune and antioxidant status, and upregulates NF- κ B/Nrf2 signalling molecules and stress resistance in *Rhynchocypris lagowskii* Dybowskii. *Aquac Nutr* 27(1):225–239. <https://doi.org/10.1111/anu.13180>
- Yu Z, Li L, Zhu R, Li M, Duan J, Wang JY, Liu YH, Wu LF (2020) Monitoring of growth, digestive enzyme activity, immune response and water quality parameters of golden crucian carp (*Carassius auratus*) in zero-water exchange tanks of biofloc systems. *Aquac Rep* 16:100283. <https://doi.org/10.1016/j.aqrep.2020.100283>

Publisher's Note

Springer Nature remains neutral with regard to jurisdictional claims in published maps and institutional affiliations. Springer Nature or its licensor (e.g., a society or other partner) holds exclusive rights to this article under a publishing agreement with the author(s) or other rightsholder(s).

Authors and affiliations

Md Aklakur¹ Md Imran Shah¹ Md Amjad¹ Udipta Roy¹ Potluri Sai Kishore² Sangeeta Mandal² Barkha Asthana¹

Corresponding author: Udipta Roy
udiptaroy@cife.edu.in

¹ Regional Research & Training Centre, ICAR-Central Institute of Fisheries Education, Motipur, Muzaffarpur, Bihar 843111, India.

² ICAR-Central Institute of Fisheries Education, Versova, Andheri West, Mumbai 400061, India.